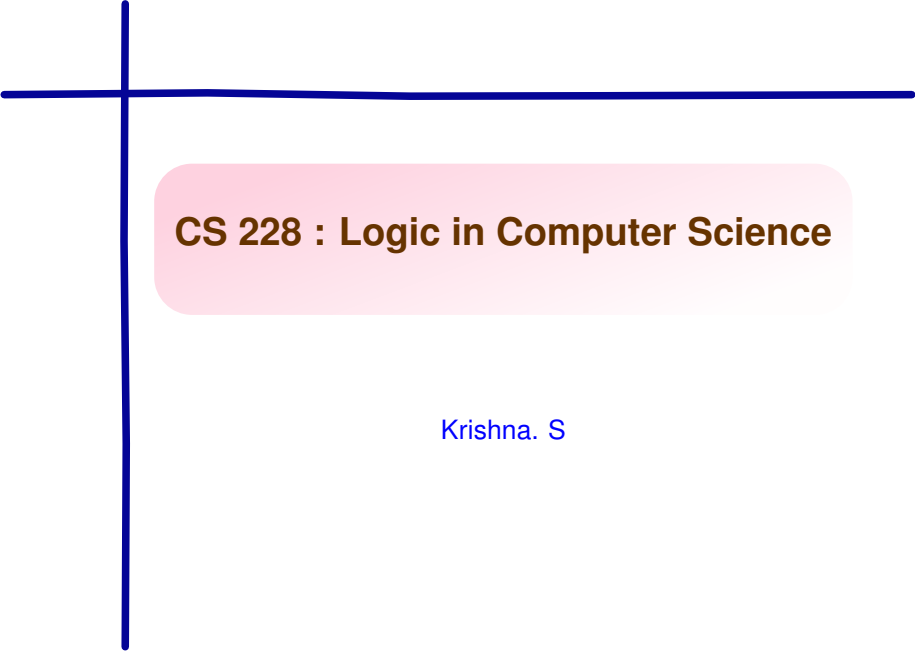




CS 228 : Logic in Computer Science

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Completeness

$$\varphi_1, \dots, \varphi_n \models \psi \Rightarrow \varphi_1, \dots, \varphi_n \vdash \psi$$

Whenever $\varphi_1, \dots, \varphi_n$ semantically entail ψ , then ψ can be proved from $\varphi_1, \dots, \varphi_n$. The proof rules are **complete**

Completeness : 3 steps

- ▶ Given $\varphi_1, \dots, \varphi_n \models \psi$
- ▶ Step 1: Show that $\models \varphi_1 \rightarrow (\varphi_2 \rightarrow (\dots (\varphi_n \rightarrow \psi) \dots))$
- ▶ Step 2: Show that $\vdash \varphi_1 \rightarrow (\varphi_2 \rightarrow (\dots (\varphi_n \rightarrow \psi) \dots))$
- ▶ Step 3: Show that $\varphi_1, \dots, \varphi_n \vdash \psi$

Completeness : Step 1

- ▶ Assume $\varphi_1, \dots, \varphi_n \models \psi$. Whenever all of $\varphi_1, \dots, \varphi_n$ evaluate to true, so does ψ .
- ▶ If $\not\models \varphi_1 \rightarrow (\varphi_2 \rightarrow (\dots (\varphi_n \rightarrow \psi) \dots))$, then ψ evaluates to false when all of $\varphi_1, \dots, \varphi_n$ evaluate to true, a contradiction.
- ▶ Hence, $\models \varphi_1 \rightarrow (\varphi_2 \rightarrow (\dots (\varphi_n \rightarrow \psi) \dots))$.

Completeness : Step 2

- ▶ Given $\models \psi$, show that $\vdash \psi$
- ▶ Assume $p_1, \dots, p_n$ are the propositional variables in ψ . We know that all the 2^n assignments of values to $p_1, \dots, p_n$ make ψ true.
- ▶ Using this insight, we have to give a proof of ψ .

Completeness : Step 2

Truth Table to Proof

Let φ be a formula with variables $p_1, \dots, p_n$. Let $\mathcal{T}$ be the truth table of φ , and let l be a line number in $\mathcal{T}$. Let $\hat{p}_i$ represent p_i if p_i is assigned true in line l , and let it denote $\neg p_i$ if p_i is assigned false in line l . Then

1. $\hat{p}_1, \dots, \hat{p}_n \vdash \varphi$ if φ evaluates to true in line l
2. $\hat{p}_1, \dots, \hat{p}_n \vdash \neg\varphi$ if φ evaluates to false in line l

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 - ▶ Assume φ evaluates to false in line l of $\mathcal{T}$. Then φ_1 evaluates to true in line l . By inductive hypothesis, $\hat{p}_1, \dots, \hat{p}_n \vdash \varphi_1$.
 - ▶ Use the $\neg i$ rule to obtain a proof of $\hat{p}_1, \dots, \hat{p}_n \vdash \neg\neg\varphi_1$.

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 - ▶ By inductive hypothesis, $\hat{q}_1, \dots, \hat{q}_k \models \varphi_1$ and $\hat{r}_1, \dots, \hat{r}_j \models \neg\varphi_2$. Then, $\hat{p}_1, \dots, \hat{p}_n \models \varphi_1 \wedge \neg\varphi_2$.

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 - ▶ By inductive hypothesis, $\hat{q}_1, \dots, \hat{q}_k \models \varphi_1$ and $\hat{r}_1, \dots, \hat{r}_j \models \neg\varphi_2$. Then, $\hat{p}_1, \dots, \hat{p}_n \models \varphi_1 \wedge \neg\varphi_2$.
 - ▶ Prove that $\varphi_1 \wedge \neg\varphi_2 \vdash \neg(\varphi_1 \rightarrow \varphi_2)$.

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- ▶ Case $\rightarrow$: $\varphi = \varphi_1 \rightarrow \varphi_2$.
 - ▶ If φ evaluates to true in line l , then there are 3 possibilities. If both φ_1, φ_2 evaluate to true, then we have $\hat{p}_1, \dots, \hat{p}_n \models \varphi_1 \wedge \varphi_2$. Proving $\varphi_1 \wedge \varphi_2 \vdash \varphi_1 \rightarrow \varphi_2$, we are done.

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 - ▶ If both φ_1, φ_2 evaluate to false, then we have $\hat{p}_1, \dots, \hat{p}_n \models \neg\varphi_1 \wedge \neg\varphi_2$. Proving $\neg\varphi_1 \wedge \neg\varphi_2 \vdash \varphi_1 \rightarrow \varphi_2$, we are done.

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 - ▶ If φ evaluates to true in line l , then there are 3 possibilities. If both φ_1, φ_2 evaluate to true, then we have $\hat{p}_1, \dots, \hat{p}_n \models \varphi_1 \wedge \varphi_2$. Proving $\varphi_1 \wedge \varphi_2 \vdash \varphi_1 \rightarrow \varphi_2$, we are done.
 - ▶ If both φ_1, φ_2 evaluate to false, then we have $\hat{p}_1, \dots, \hat{p}_n \models \neg\varphi_1 \wedge \neg\varphi_2$. Proving $\neg\varphi_1 \wedge \neg\varphi_2 \vdash \varphi_1 \rightarrow \varphi_2$, we are done.
 - ▶ Last, if φ_1 evaluates to false and φ_2 evaluates to true, then we have $\hat{p}_1, \dots, \hat{p}_n \models \neg\varphi_1 \wedge \varphi_2$. Proving $\neg\varphi_1 \wedge \varphi_2 \vdash \varphi_1 \rightarrow \varphi_2$, we are done.

Truth Table to Proof

- ▶ Prove the cases $\wedge, \vee$.

On An Example

We know $\models (p \wedge q) \rightarrow p$. Using this fact, show that $\vdash (p \wedge q) \rightarrow p$.

- ▶ $p, q \vdash (p \wedge q) \rightarrow p$
- ▶ $\neg p, q \vdash (p \wedge q) \rightarrow p$
- ▶ $p, \neg q \vdash (p \wedge q) \rightarrow p$
- ▶ $\neg p, \neg q \vdash (p \wedge q) \rightarrow p$

Now, combine the 4 proofs above to give a single proof for $\vdash (p \wedge q) \rightarrow p$.

Completeness : Steps 2, 3

- ▶ Step 2: From $\models \varphi_1 \rightarrow (\varphi_2 \rightarrow (\dots(\varphi_n \rightarrow \psi)\dots))$, use LEM on all the propositional variables of $\varphi_1, \dots, \varphi_n, \psi$ to obtain $\vdash \varphi_1 \rightarrow (\varphi_2 \rightarrow (\dots(\varphi_n \rightarrow \psi)\dots))$.

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- ▶ Step 3: Take the proof $\vdash \varphi_1 \rightarrow (\varphi_2 \rightarrow (\dots (\varphi_n \rightarrow \psi) \dots))$. This proof has n nested boxes, the i th box opening with the assumption φ_i . The last box closes with the last line ψ . Hence, the line immediately after the last box is $\varphi_n \rightarrow \psi$.

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- ▶ In a similar way, the $(n - 1)$ th box has as its last line $\varphi_n \rightarrow \psi$, and hence, the line immediately after this box is $\varphi_{n-1} \rightarrow (\varphi_n \rightarrow \psi)$ and so on.

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- ▶ In a similar way, the $(n - 1)$ th box has as its last line $\varphi_n \rightarrow \psi$, and hence, the line immediately after this box is $\varphi_{n-1} \rightarrow (\varphi_n \rightarrow \psi)$ and so on.
- ▶ Add premises $\varphi_1, \dots, \varphi_n$ on the top. Use MP on the premises, and the lines after boxes 1 to n in order to obtain ψ .

Summary

Propositional Logic is sound and complete.

Satisfiability and Validity

A formula φ over variables $\{p_1, \dots, p_n\}$ is satisfiable if there exists a valuation for $p_1, \dots, p_n$ which makes φ true.

A formula φ over variables $\{p_1, \dots, p_n\}$ is unsatisfiable if there does not exist any valuation for $p_1, \dots, p_n$ which makes φ true.

A formula φ over variables $\{p_1, \dots, p_n\}$ is valid if φ is true for all valuations to $p_1, \dots, p_n$.

Checking Satisfiability

- ▶ The SAT problem : Given a formula φ , is it satisfiable?
- ▶ Given a formula φ , is it valid?
- ▶ SAT is NP-complete. NP represents non-deterministic polynomial time.
- ▶ Given a witness, it is easy to check if the witness is a valid witness in polynomial time, Finding a witness is not as easy.
 - ▶ Given a valuation α for the variables $p_1, \dots, p_n$ of a formula φ , we can check if φ evaluates to true under α in time polynomial in n .
 - ▶ Finding whether such a valuation α exists is not as easy.
- ▶ SAT solvers are tools which implement heuristics to check satisfiability.
- ▶ The input to a SAT solver is a formula in some specified form.